OO-S-256/16A
June 13, 1986
SUPERSEDING
OO-S-00256(GL)/16
June 11, 1985

FEDERAL SPECIFICATION SHEET

SEWING MACHINE, INDUSTRIAL; DARNING, CYLINDER BED, SINGLE NEEDLE, 301 STITCH, PARACHUTE REPAIR

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring the sewing machines described herein shall consist of this document and the issue effect of Specification OO-S-256.

1. SCOPE

1.1 Scope. This specification covers the specific requirements for single needle, cylinder bed, 301 stitch, heavy duty, darning, industrial sewing machines used in the fabrication and repair of aerial delivery equipment.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

6-1-9367 Sewing Machine, Industrial, Darning, Single Needle, Heavy Duty: Parachute Repair and Fabrication Test Requirements

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

AMSC N/A FSC 3530

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3. REQUIREMENTS

3.1 Machine, heavy duty. The heavy duty darning machine shall have the capability of darning light to heavy cotton and nylon duck utilizing metric needle sizes 90, 110, 120, 125, 130, 140, and 180. The machine shall operate at speeds not less than 2000 or greater than 3000 stitches per minute. The machine shall be manually or semi-automatically lubricated. The machine shall be furnished with a cone on the end of the cylinder bed. The machine shall be table mounted. Each machine shall be supplied with a bobbin winder and knee lifter.

3.1.1 Machine feed. The machine shall not have a type of feeding mechanism.

3.1.2 Machine height under the presser foot. The machine shall provide a minimum height of 3/8 of an inch under the presser foot.

3.1.3 Machine, working room to the right of the needle. The machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 16 inches maximum.

3.1.4 Machine, table top. The machine shall be furnished with a type II table top. The outside dimensions of the table top shall be 20 by 40 by 2 inches thick with a tolerance of ± 1/4 of an inch for each dimension.

3.1.5 Spare parts. The machine shall be furnished with the following spare parts in the quantity indicated.

Spare parts

Item identification	Quantity
Face plate screws	2
Thread take up spring	1
Thread tension spring thumb nut	1
Tension stud	1
Sewing hook assembly	1
Sewing hook holding screw	1
Bobbin, metal	10
Bobbin case assembly	1
Saddle gear screw	1
Needle bar thread guide	1
Needle bar thread guide screw	2
Needle set screw	2
Lifting bracket screw	2
Presser foot finger guard spring	1
Presser foot holding screw	1
Presser foot	1

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Spare parts (cont'd)

Item identification	Quantity
Presser foot finger guard spring screw	2
Thread guide screw	2
Needle brushing thread guide	1
Top plate holding screw	1
Take up lever stud	1
Needle, round point, chrome finish:	
Metric sizes:	
90	100
110	100
125	100
140	100
180	100

3.2 Performance.

3.2.1 Machine operation. The heavy duty machine shall be capable of darning and mending panels, pack sections, and deployment bags of personnel, extraction, and cargo parachutes when tested as specified in 4.1.1 and 4.2.1. The machine shall show no signs of stalling, pounding, shearing of threads, failure to lock stitches, or deformation of any part during the darning and mending of these articles.

4. QUALITY ASSURANCE PROVISIONS

4.1 First article testing. When a first article is required it shall be tested in accordance with the applicable test method specified in table I.

TABLE I. First article tests

Sewing machine	Requirement paragraph	Test method
Heavy duty	3.2.1	4.1.1

4.1.1 Machine test. The heavy duty machine shall be operated to darn and mend all of the typical parachute details as shown on Drawing 6-1-9367. Any nonconformance with the requirements of 3.2.1 shall constitute failure of this test.

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4.2 End item testing. Each end item shall be tested in accordance with the applicable test method specified in table II. Failure of any test shall be cause for rejection of the end item involved.

TABLE II. End item testing

Sewing machine	Requirement paragraph	Test method
Heavy duty	3.2.1	4.2.1

4.2.1 Machine test. Each machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9367. Any nonconformance with the requirements of 3.2.1 shall constitute failure of this test. The completed test sample shall be supplied with each machine furnished.

5. PACKAGING

Packaging shall be as specified in the basic specification and in the contract or purchase order.

6. NOTES

6.1 Intended use. The sewing machines are intended for industrial use for fabricating and repairing parachutes.

6.2 Ordering data.

Title, number, and date of this specification.

MILITARY INTERESTS:

Custodians:

Army - GL
Air Force - 99

Review activity:

DLA-GS

User activity:

Air Force - 50

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - GL

Project No. 3550-0081

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